Divisibilities — Solutions

Problem 1. (Spain MO 1996 Problem 1) The natural numbers a and b are such that $\dfrac{a+1}{b} + \dfrac{b+1}{a}$ is an integer. Show that the greatest common divisor of a and b is not greater than $\sqrt{a+b}$.

Solution. Let $a = dm$ and $b = dn$ where m and n are positive integers with $(m, n) = 1$. We have

$$\frac{a+1}{b} + \frac{b+1}{a} = \frac{dm+1}{dn} + \frac{dn+1}{dm} = \frac{dm^2 + dn^2 + m + n}{dmn}.$$

Since this is an integer, we have $d \mid dm^2 + dn^2 + m + n$. This implies $d \mid m+n$, and hence $d \leqslant m+n$. Thus, we have $d \leqslant \sqrt{d(m+n)} = \sqrt{a+b}$. □

Problem 2. (Iran MO 3rd Round 2003) Does there exist an infinite set of integers S such that for every $a, b \in S$ we have $a^2 + b^2 - ab \mid (ab)^2$?

Solution. No. We fix any nonzero $a \in S$ (if there is not such a, then $|S| \leqslant 1$). For any $b \in S$ different from a, we have

$$(ab)^2 = a^2(a^2 + b^2 - ab) - a^3(a - b),$$

so that $a^2 + b^2 - ab \mid a^3(a - b)$. Since $a^2 + b^2 - ab > 0$, this implies $a^2 + b^2 - ab \leqslant |a^3(a - b)|$. As $a^2 + b^2 - ab$ is a degree 2 polynomial in b while $a^3(a - b)$ has degree 1, the inequality only holds for finitely many b's (recall that a is fixed). This means S is a finite set. □

Problem 3. (Germany MO 2000 Problem 4) Determine all positive integers n for which there exists a positive integer d with the property that n is divisible by d and $n^2 + d^2$ is divisible by $d^2n + 1$.

Solution. n can be any positive integer of the form m^4 where m is a positive integer.

Let $n = dk$. We need $d^3k + 1 \mid d^2k^2 + d^2$. Clearly, $(d^3k + 1, d) = (1, d) = 1$. Thus, we need $d^3k + 1 \mid k^2 + 1$. This easily implies $d^3k + 1 \leqslant k^2 + 1$, and hence $d^3 \leqslant k$.

On the other hand, note that $d^3k + 1 \mid d^6(k^2 + 1) = (d^3k + 1)(d^3k - 1) + (d^6 + 1)$. This implies $d^3k + 1 \mid d^6 + 1$. Thus, we have $d^3k + 1 \leqslant d^6 + 1$, and hence $k \leqslant d^3$.

Combining the two inequalities, we must have $k = d^3$, and hence $n = d^4$. Conversely, if $n = m^4$, then we can take $d = m$ so that $n^2 + d^2 = m^8 + m^2$ is divisible by $d^2n + 1 = m^6 + 1$. □

Problem 4. Find all positive integers n satisfying the following condition: For any positive integers a and b, if $n \mid a^2b + 1$, then $n \mid a^2 + b$.

Solution. n can be any positive divisor of 240, i.e. $1, 2, 3, 4, 5, 6, 8, 10, 12, 15, 16, 20, 24, 30, 40, 48, 60, 80, 120, 240$.

We claim that n satisfies the condition if and only if $n \mid k^4 - 1$ for any positive integer k relatively prime to n.

Firstly, suppose $n \mid k^4 - 1$ for any positive integer k relatively prime to n. If $n \mid a^2b + 1$, clearly $(n, a) = 1$. Since $n \mid a^4 - 1$, we have

$$0 \equiv a^2(a^2b + 1) = a^4b + a^2 \equiv b + a^2 \pmod{n}.$$

So n satisfies the condition.

Secondly, suppose n satisfies the condition. For any positive integer k relatively prime to n, there exists a positive integer b such that $n \mid k^2b + 1$. This implies $n \mid k^2 + b$, or $b \equiv -k^2 \pmod{n}$. It follows that

$$0 \equiv k^2b + 1 \equiv -k^4 + 1 \pmod{n}.$$

Thus, $n \mid k^4 - 1$.

Note that it suffices to consider $n = p^s$ for some prime p. Indeed, if $(m_1, m_2) = 1$, then $n = m_1m_2$ is a solution if and only if both $n = m_1$ and $n = m_2$ are solutions. So we check for prime powers as follows.

For any p^s where p is an odd prime, we consider $k = 2$. This requires $p^s \mid 15$. Thus, p^s can only be 3 and 5. Conversely, by the Fermat little theorem, we have $3 \mid k^4 - 1$ whenever $3 \nmid k$ and $5 \mid k^4 - 1$ whenever $5 \nmid k$.

Lastly, for $p = 2$, we can prove that $16 \mid k^4 - 1 = (k^2 + 1)(k^2 - 1)$ for any odd k, since $k^2 + 1$ is even, and it is well-known that $8 \mid k^2 - 1$. However, we have $32 \nmid 3^4 - 1 = 80$.

Therefore, n can be any one of $2^a3^b5^c$ with $0 \leqslant a \leqslant 4$, $b = 0, 1$ and $c = 0, 1$, which means n is any positive divisor of 240. $\square$

Problem 5. (All-Russian MO 2001 Grade 10 Problem 8) Find all odd positive integers $n > 1$ such that if a and b are relatively prime divisors of n, then $a + b - 1$ divides n.

Solution. n can be any power of an odd prime.

If $n = p^k$ for some odd prime p and positive integer k, the only relatively prime divisors of n are 1 and p^s for some $s \leqslant k$. We have $1 + p^s - 1 = p^s \mid p^k$. So n satisfies the condition.

In the following, we suppose n has at least 2 different prime divisors. Let $n = p^km$ where p is the smallest prime divisor of n, and k, m are positive integers such that $p \nmid m$ and $m > 1$. Consider $a = m$ and $b = p$. Then we have $m + p - 1 \mid p^km$. Since $m + p - 1 > m$, it is impossible that $m + p - 1 \mid m$. Thus, $p \mid m + p - 1$, and hence $p \mid m - 1$. Let $m = p^tc + 1$ where $t \geqslant 1$ and $p \nmid c$. This gives $p^tc + p \mid p^km$, and hence $p^{t-1}c + 1 \mid p^{k-1}m$. Note that $p \nmid p^{t-1}c + 1$. So we have

$$p^{t-1}c + 1 \mid m = p^tc + 1 = p(p^{t-1}c + 1) - (p - 1).$$

This implies $p^{t-1}c+1 \mid p-1$. Thus, $p^{t-1} < p^{t-1}c+1 \leqslant p-1 < p$. The only possibility is $t=1$. This becomes $c+1 \mid p-1$, so that $c \leqslant p-2$. Now, we have $m = pc+1 \leqslant p^2-2p+1 < p^2$. Recall that p is the smallest prime divisor of n. Therefore, $m = q$ is another prime divisor of n.

If $k \geqslant 2$, consider $a = p^2$ and $b = q$. This implies $p^2+q-1 \mid p^k q$. We have shown that $p \parallel q-1$, and so $p \parallel p^2+q-1$. Also, note that $(p^2+q-1, q) = (p^2-1, q) = 1$ since $p^2-1 = (p-1)(p+1)$, while $q \geqslant p+2$. Therefore, we have $p^2+q-1 = p$. Clearly, this is impossible as $p^2+q-1 > p^2 > p$.

It remains to consider $n = pq$. Consider $a = p$ and $b = q$. This implies $p+q-1 \mid pq$. Again, $q \nmid p+q-1$. So we must have $p+q-1 = p$. This is impossible.

Therefore, n can only be p^k. $\square$

Problem 6. (Turkey MO 2015 Problem 6) Find all positive integers n such that for any positive integer a relatively prime to n, $2n^2 \mid a^n - 1$.

Solution. n can be $2, 6, 42, 1806$.

If n is odd, we can take $a = 2$ so that $a^n - 1$ is odd, and thus $2n^2 \nmid a^n - 1$. So n is even. Let $n = 2^k m$ where m is odd. We first prove that $k = 1$. Indeed, by the Chinese remainder theorem, there exists a such that $a \equiv 1 \pmod m$ and $a \equiv 3 \pmod 8$. Clearly, $(a, n) = 1$. We have $v_2(2n^2) = 2k+1$ and

$$v_2(a^n - 1) = v_2(a^2-1) + v_2(n) - 1 = v_2(a+1) + v_2(a-1) + k - 1 = k+2$$

by the lifting the exponent lemma. Therefore, we have $2k+1 \leqslant k+2$, and so $k = 1$.

Similarly, we can prove that n is squarefree. Indeed, for any odd prime divisor p of n, let $n = p^k c$ where $p \nmid c$. There exists a such that $a \equiv 2 \pmod p$ and $a \equiv 1 \pmod c$ by the Chinese remainder theorem. Clearly, $(a, n) = 1$. We have $v_p(2n^2) = 2k$ and

$$v_p(a^n - 1) = v_p(a-1) + v_p(n) = k$$

by the lifting the exponent lemma. Therefore, we have $2k \leqslant k$, and so $k = 1$.

Let $n = p_1 p_2 \cdots p_s$ where $2 = p_1 < p_2 < \cdots < p_s$. If $(a, n) = 1$, a must be odd, and hence $8 \mid a^n - 1$. Therefore, $2n^2 \mid a^n - 1$ holds if and only if $p_j^2 \mid a^n - 1$ for each $j \geqslant 2$. The divisibility holds if $\varphi(p_j^2) \mid n$. Conversely, by choosing a to be a primitive root modulo p_j^2, we need $\varphi(p_j^2) \mid n$. Therefore, it remains to find all $n = p_1 p_2 \cdots p_s$ such that $\varphi(p_j^2) \mid n$ for each $j \geqslant 2$. Since $\varphi(p_j^2) = p_j(p_j - 1)$, this means $p_j - 1 \mid n$ for each $j \geqslant 2$.

When $s = 1$, we must have $n = 2$. When $s \geqslant 2$, since $p_2 - 1 \mid p_1 p_2 \cdots p_s$, and $p_2 - 1 < p_2, \ldots, p_s$, we must have $p_2 - 1 \mid p_1 = 2$. The only possibility is $p_2 = 3$, which gives another solution $n = 2 \times 3 = 6$.

When $s \geqslant 3$, similarly, we have $p_3 - 1 \mid p_1 p_2 = 6$. The only possibility is $p_3 = 7$, which gives another solution $n = 2 \times 3 \times 7 = 42$.

When $s \geqslant 4$, we have $p_4 - 1 \mid p_1 p_2 p_3 = 42$. The only possibility is $p_4 = 43$, which gives another solution $n = 2 \times 3 \times 7 \times 43 = 1806$.

Lastly, when $s \geqslant 5$, we have $p_5 - 1 \mid p_1p_2p_3p_4 = 1806$. Since $p_5 > 43$, the only possibilities are $p_5 - 1 = 86, 129, 258, 301, 602, 903, 1806$. But then none of the solution p_5 is a prime. Therefore, n can only be $2, 6, 42, 1806$. □

Problem 7. (Kazakhstan MO 2019 Problem 4) Find all positive integers $n, k, a_1, a_2, \ldots, a_k$ such that $n^{k+1} + 1$ is divisible by $(na_1 + 1)(na_2 + 1)\cdots(na_k + 1)$.

Solution. The answers are given as follows.

- $k = 1$ and $n = a_1$;
- $k = 2$ and $(a_1, a_2) = (1, n-1), (n-1, 1)$ where $n \geqslant 2$

Let $n^{k+1} + 1 = c(na_1+1)(na_2+1)\cdots(na_k+1)$ for some positive integer c. By considering modulo n, we obtain $c \equiv 1 \pmod{n}$. Since

$$c = \frac{n^{k+1}+1}{(na_1+1)(na_2+1)\cdots(na_k+1)} \leqslant \frac{n^{k+1}+1}{(n+1)^k} < n+1,$$

we must have $c = 1$. Thus, we have

$$n^{k+1} + 1 = (na_1+1)(na_2+1)\cdots(na_k+1) = a_1\cdots a_k n^k + \cdots + (a_1 + \cdots + a_k)n + 1,$$

and hence $n^k = a_1 \cdots a_k n^{k-1} + \cdots + (a_1 + \cdots + a_k)$.

If $k = 1$, the equation becomes $n^2 + 1 = na_1 + 1$. Clearly, the solution is $n = a_1$.

If $k = 2$, the equation becomes $n^2 = a_1a_2n + a_1 + a_2$. Note that $n \mid a_1 + a_2$. If $a_1 + a_2 \geqslant 2n$, since $a_1a_2 \geqslant a_1 + a_2 - 1$ (which is equivalent to $(a_1-1)(a_2-1) \geqslant 0$), we have

$$n^2 = a_1a_2n + a_1 + a_2 \geqslant (2n-1)n + (2n) = 2n^2 + n,$$

contradiction. This shows $a_1 + a_2 = n$. If $a_1, a_2 \geqslant 2$, then we have $a_1a_2 \geqslant 2(n-2)$. This yields

$$n^2 = a_1a_2n + a_1 + a_2 \geqslant (2n-4)n + n = 2n^2 - 3n.$$

So we must have $n \leqslant 3$. But then $a_1 + a_2 \leqslant 3$, and so it is impossible that $a_1, a_2 \geqslant 2$. Therefore, we must have $(a_1, a_2) = (1, n-1), (n-1, 1)$. In that case we check that

$$a_1a_2n + a_1 + a_2 = (n-1)n + n = n^2.$$

If $k \geqslant 3$, clearly $n > 1$. Let s be the number of a_j's equal to 1. Note that

$$\begin{aligned} n^{k+1} + 1 &= \big((n+1) - 1\big)^{k+1} + 1 \\ &= 1 + (-1)^{k+1} + (-1)^k(k+1)(n+1) + (-1)^{k-1}\frac{(k+1)k}{2}(n+1)^2 + (n+1)^3 t \end{aligned}$$

for some $t \in \mathbb{Z}$ by the binomial theorem.

If $s \geqslant 1$, then $n+1 \mid n^{k+1}+1$, and hence $n+1 \mid 1+(-1)^{k+1}$. This implies k is even. If $s \geqslant 2$, then $(n+1)^2 \mid n^{k+1}+1$. This yields $(n+1)^2 \mid (k+1)(n+1)$, so that $n+1 \mid k+1$. If $s \geqslant 3$, then $(n+1)^3 \mid n^{k+1}+1$. Note that $n+1$ divides $\dfrac{(k+1)k}{2}$ since k is even and $n+1 \mid k+1$. Thus, we have $(n+1)^2 \mid k+1$. This implies

$$k \geqslant n^2 + 2n.$$

On the other hand, by the binomial theorem, we have

$$n^{k+1}+1 = \prod_{j=1}^{k}(na_j+1) \geqslant (n+1)^k \geqslant kn^{k-1}+1 \geqslant (n^2+2n)n^{k-1}+1.$$

This is a contradiction. Therefore, at most 2 of the a_j's are equal to 1, while the others are at least 2.

Recall that $n^k = a_1 \cdots a_k n^{k-1} + \cdots + (a_1 + \cdots + a_k)$. Clearly, this implies $n^k > a_1 \cdots a_k n^{k-1}$, and so

$$a_1 \cdots a_k < n. \tag{1}$$

Also, the equation implies $n \mid a_1 + \cdots + a_k$, and so

$$a_1 + \cdots + a_k \geqslant n. \tag{2}$$

Combining (1) and (2), we get $a_1 + \cdots + a_k > a_1 \cdots a_k$. However, this only holds for some small cases. Indeed, whenever $a_1 \cdots a_k \geqslant a_1 + \cdots + a_k$, the following relations hold:

$$a_1 \cdots a_{k-1}(a_k+1) \geqslant a_1 + \cdots + a_{k-1} + (a_k+1),$$
$$a_1 \cdots a_k(2) \geqslant a_1 + \cdots + a_k + 2.$$

Therefore, using the base case $(1)(1)(2)(2)(2) \geqslant 1+1+2+2+2$, we can prove by induction that $a_1 \cdots a_k \geqslant a_1 + \cdots + a_k$ for $k \geqslant 5$. This contradicts with (1) and (2).

Similarly, for $k=4$, we use the base cases $(1)(1)(2)(4) \geqslant 1+1+2+4$, $(1)(1)(3)(3) \geqslant 1+1+3+3$ and $(1)(2)(2)(2) \geqslant 1+2+2+2$. Therefore, the only possibility for $a_1a_2a_3a_4 < a_1+a_2+a_3+a_4$ to hold is $(a_1,a_2,a_3,a_4) = (1,1,2,2), (1,1,2,3)$.

If $(a_1,a_2,a_3,a_4) = (1,1,2,2)$, (1) becomes $n > 4$. Also, recall that $n \mid a_1+a_2+a_3+a_4 = 6$. Therefore, we must have $n=6$. But we check that $6^5+1 \neq (6+1)^2(12+1)^2$.

If $(a_1,a_2,a_3,a_4) = (1,1,2,3)$, (1) becomes $n > 6$. Also, recall that $n \mid a_1+a_2+a_3+a_4 = 7$. Therefore, we must have $n=7$. But we check that $7^5+1 \neq (7+1)^2(14+1)(21+1)$.

Lastly, for $k=3$, none of the a_j's can be 1 since k is odd. Thus, we can use the same argument with the base case $(2)(2)(2) \geqslant 2+2+2$ to obtain a contradiction. $\square$

Problem 8. (China TST 2009 Problem 3) Prove that for any odd prime number p, the number of positive integer n satisfying $p \mid n! + 1$ is less than or equal to $cp^{\frac{2}{3}}$, where c is a constant independent of p.

Solution. If $p \mid n! + 1$, clearly $n \leqslant p - 1$. Suppose $a_1 < a_2 < \cdots < a_k$ are the only positive integers for which $p \mid a_j! + 1$. For each $j = 1, 2, \ldots, k-1$, we have

$$p \mid (a_{j+1}! + 1) - (a_j! + 1) = a_j![(a_j + 1)(a_j + 2) \cdots (a_{j+1}) - 1],$$

so that $p \mid (a_j + 1)(a_j + 2) \cdots (a_{j+1}) - 1$. Define $b_j = a_{j+1} - a_j$ for each j. This implies

$$(a_j + 1)(a_j + 2) \cdots (a_j + b_j) \equiv 1 \pmod{p}.$$

For each fixed $d \geqslant 1$, there are at most d solutions to

$$(x + 1)(x + 2) \cdots (x + d) \equiv 1 \pmod{p}.$$

As $1 \leqslant a_j \leqslant p - 1$, this means at most d of the b_j's are equal to d.

For each positive integer d, suppose there are $c_d \leqslant d$ of the b_j's equal to d. Then we have

$$c_1 + c_2 + c_3 + \cdots = k - 1 \tag{1}$$

and

$$c_1 + 2c_2 + 3c_3 + \cdots = b_1 + b_2 + \cdots + b_{k-1} = a_k - a_1 \leqslant p - 2. \tag{2}$$

We are going to prove that $k \ll p^{\frac{2}{3}}$ (which means $k \leqslant cp^{\frac{2}{3}}$ for some constant c) under the constraints (1), (2) and $c_d \leqslant d$ for all d.

For a fixed k, the sum $c_1 + c_2 + c_3 + \cdots$ is fixed. If there exist $i < j$ such that $c_i < i$ and $c_j \geqslant 1$, we can replace c_i and c_j by $c_i + 1$ and $c_j - 1$. All constraints still hold, and the value of $c_1 + 2c_2 + 3c_3 + \cdots$ is decreased. Repeating the same process, we find that $c_1 + 2c_2 + 3c_3 + \cdots$ is minimized when $c_j = j$ for $j = 1, 2, \ldots, m$, and c_{m+1} is the only possible nonzero term such that $c_{m+1} < m + 1$.

In that case, (2) becomes

$$p \geqslant 2 + c_1 + 2c_2 + 3c_3 + \cdots > 1^2 + 2^2 + \cdots + m^2 = \frac{m(m+1)(2m+1)}{6} > \frac{m^3}{3}.$$

Lastly, by equation (1), we obtain

$$k = 1 + 2 + \cdots + m + c_{m+1} + 1 \leqslant 1 + 2 + \cdots + m + (m + 1) = \frac{(m+1)(m+2)}{2} \ll m^2 \ll p^{\frac{2}{3}}.$$

$\square$